

KHOJA.

Smart Tourism and Bus Navigation System

Today's Agenda

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Introduction

- **Smart travel opportunities:** The rapid growth of urban transportation and digital mapping has created a need for more connected and smarter travel experiences
- **Integrated platform:** KHOJA is designed to bridge the gap between fragmented tourism and transport systems by combining **real-time tracking, interactive navigation, and community sharing.**
- **Role-based access:** The system features specific portals for **users, drivers, and administrators to manage their respective tasks.**

Problem Statement

● **PROBLEM 1**

Lack of real-time data:
Commuters often face difficulties because there is no accurate information regarding the current location of buses or their arrival times.

● **PROBLEM 2**

Fragmented services: Existing solutions for navigation, booking, and tracking are often separate, leading to a disconnected user experience.

● **PROBLEM 3**

Limited community input:
Most platforms do not allow users to contribute local knowledge, such as new bus stops or road conditions

Objectives

Enable users, drivers, and administrators to manage transportation information.

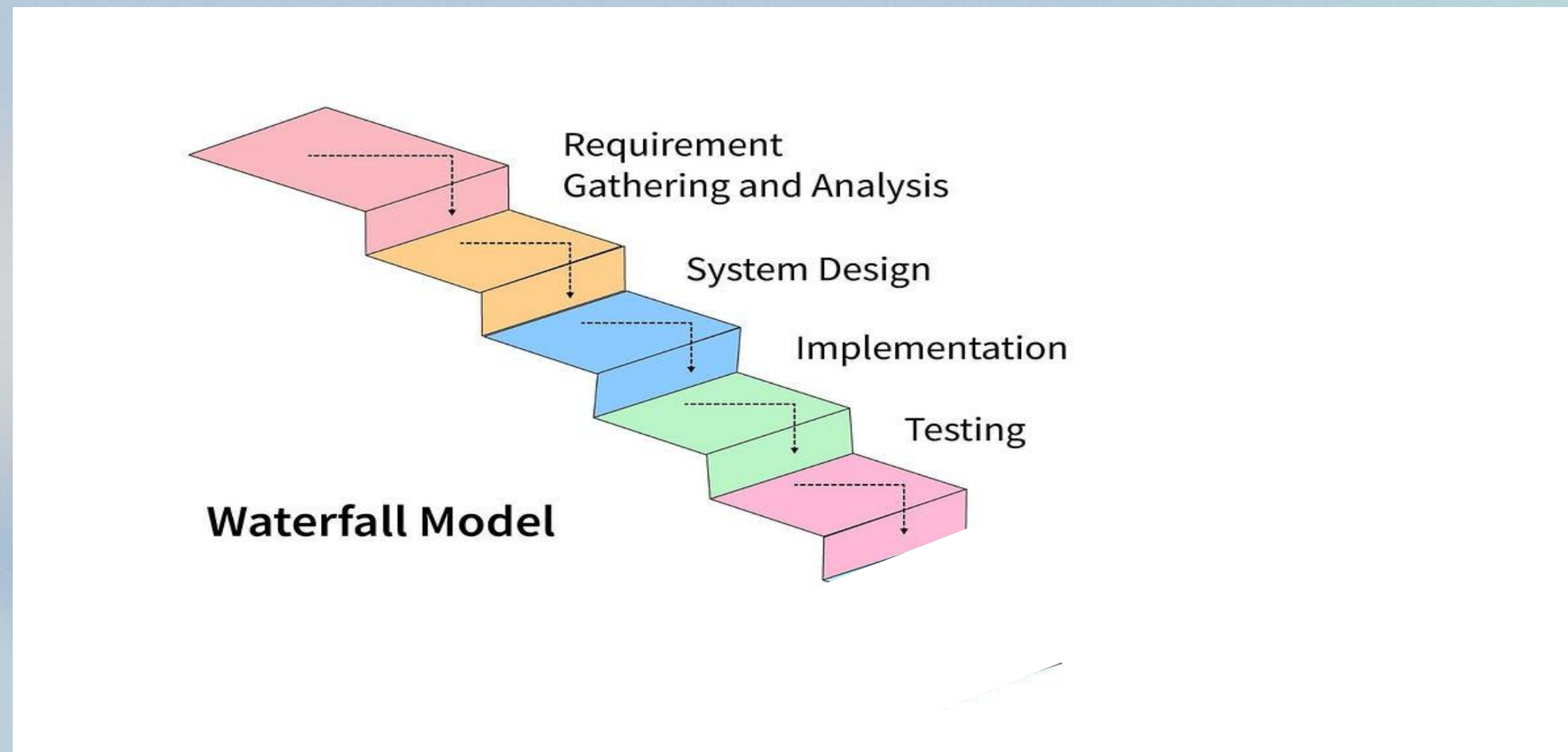
Develop a web-based bus management system.

Implement interactive maps for routes, locations, and real-time bus tracking.

Methodology

Waterfall Software Development Methodology

This methodology is suitable for developing the **Smart Tourism Navigation and Bus Management System** as the requirements, features, and roles in this system are fixed and clearly defined from the beginning. The Waterfall Model also provides a systematic and structured approach to software development, which allows for proper planning, clear documentation, and better scheduling of the entire project.



Scope



Limitation

Internet Dependency

Gps Accuracy

Manual Verification Process

Limited Multi Bus Route Calculation

Platform Availibility

Existing System

Google Maps

Trip Advisor

Tech Stack

Waterfall Software Development Methodology

01 Frontend

HTML, CSS, JAVASCRIPT

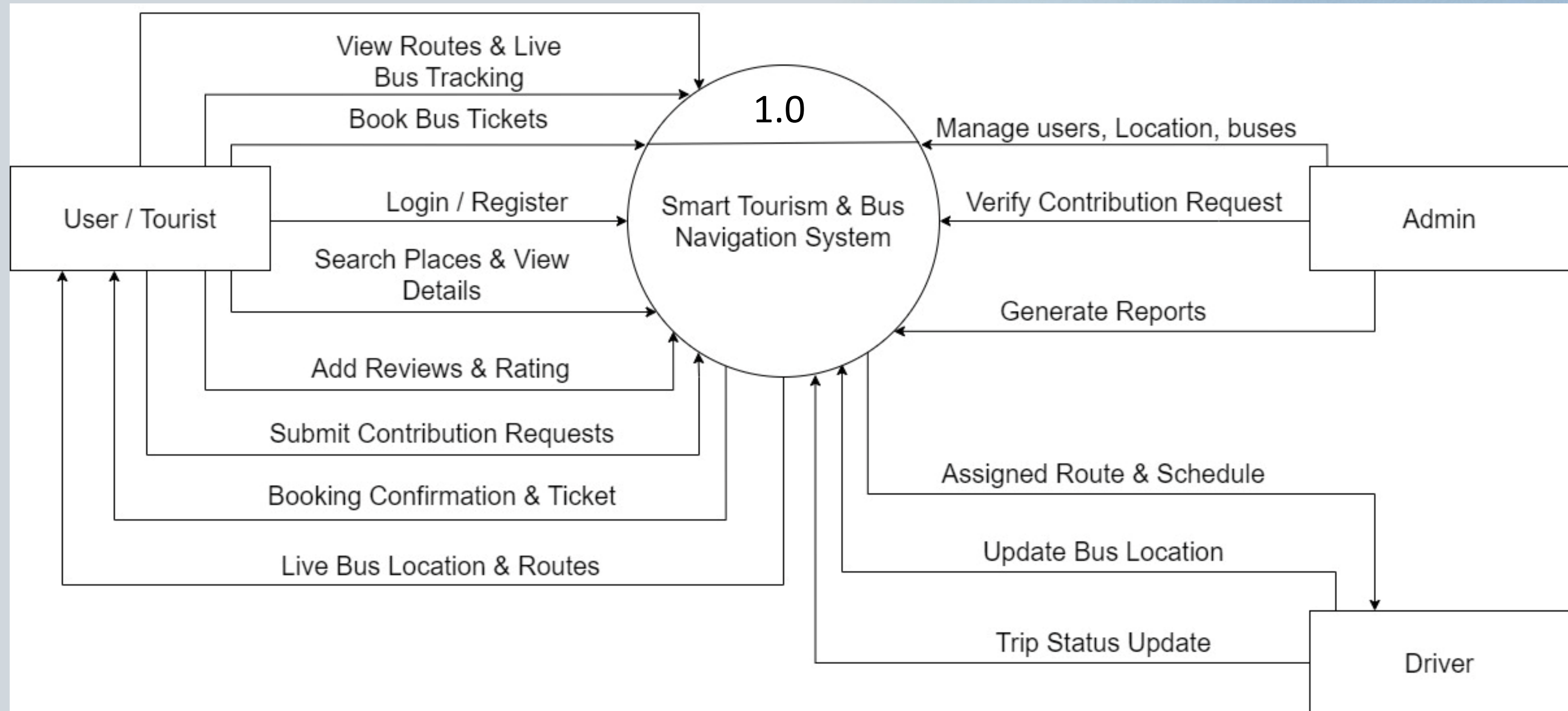
02 Backend

PHP, JAVA

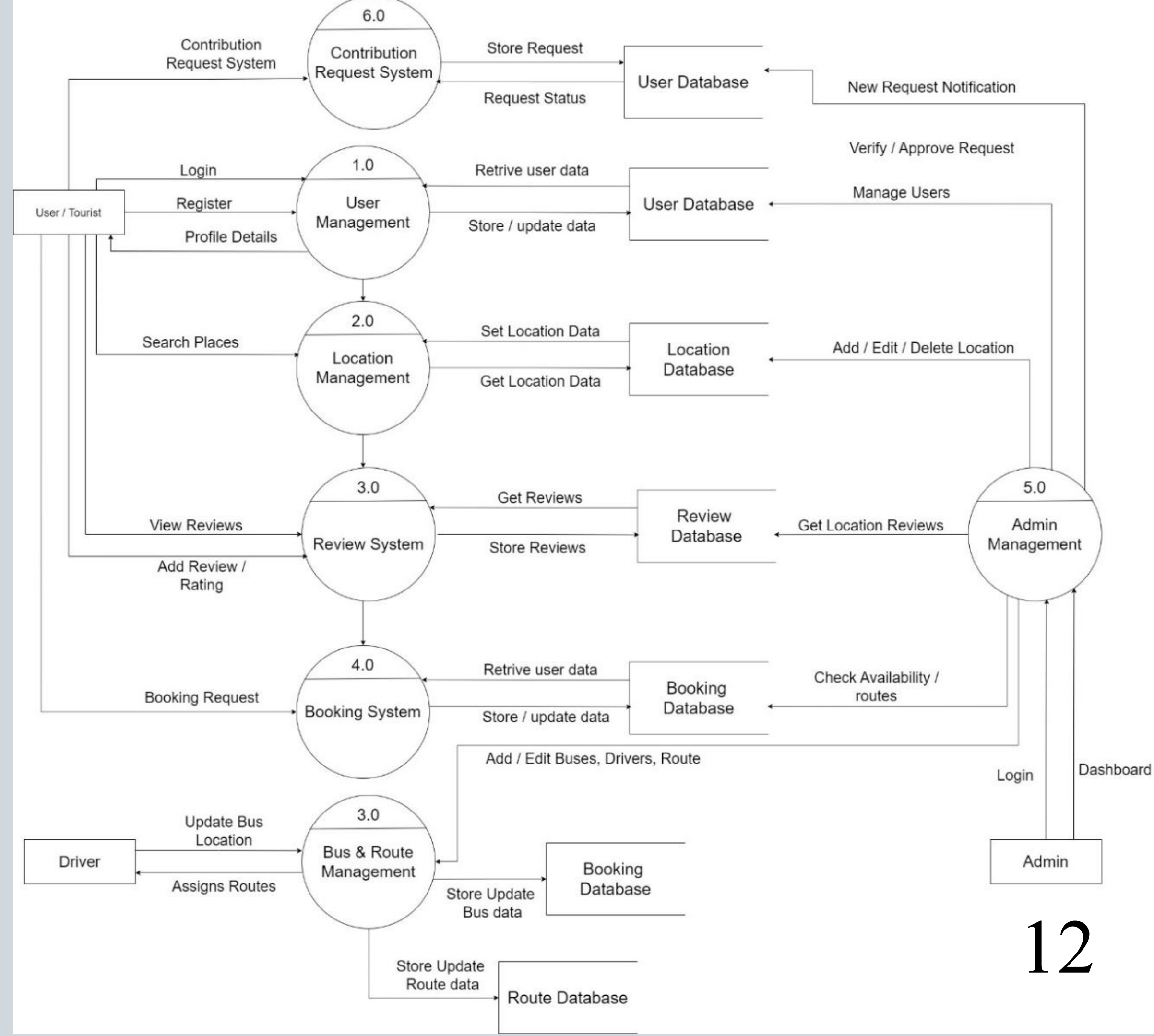
03 Database

MYSQL

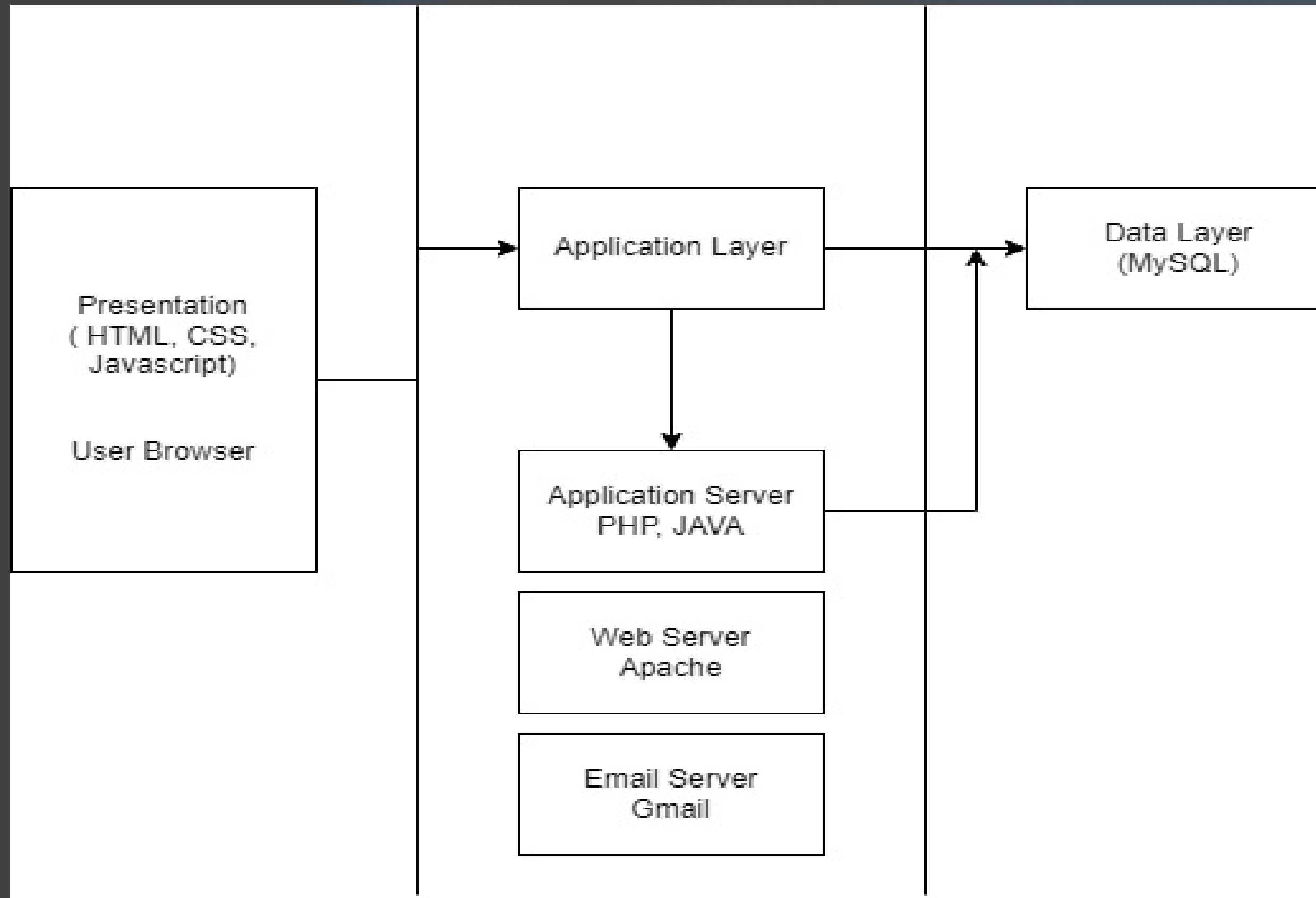
Context Diagram



DFD level 1

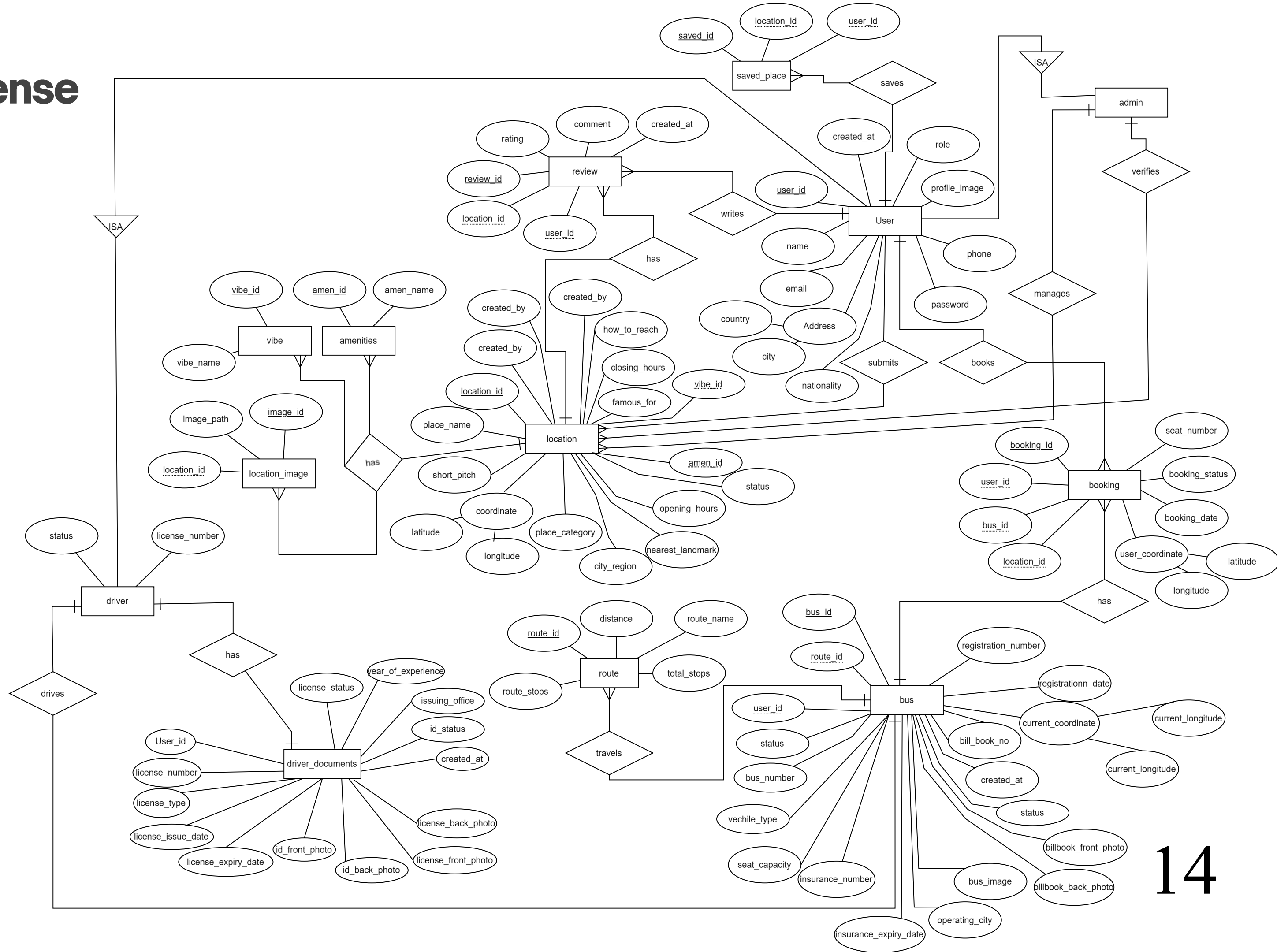


System Architecture Date

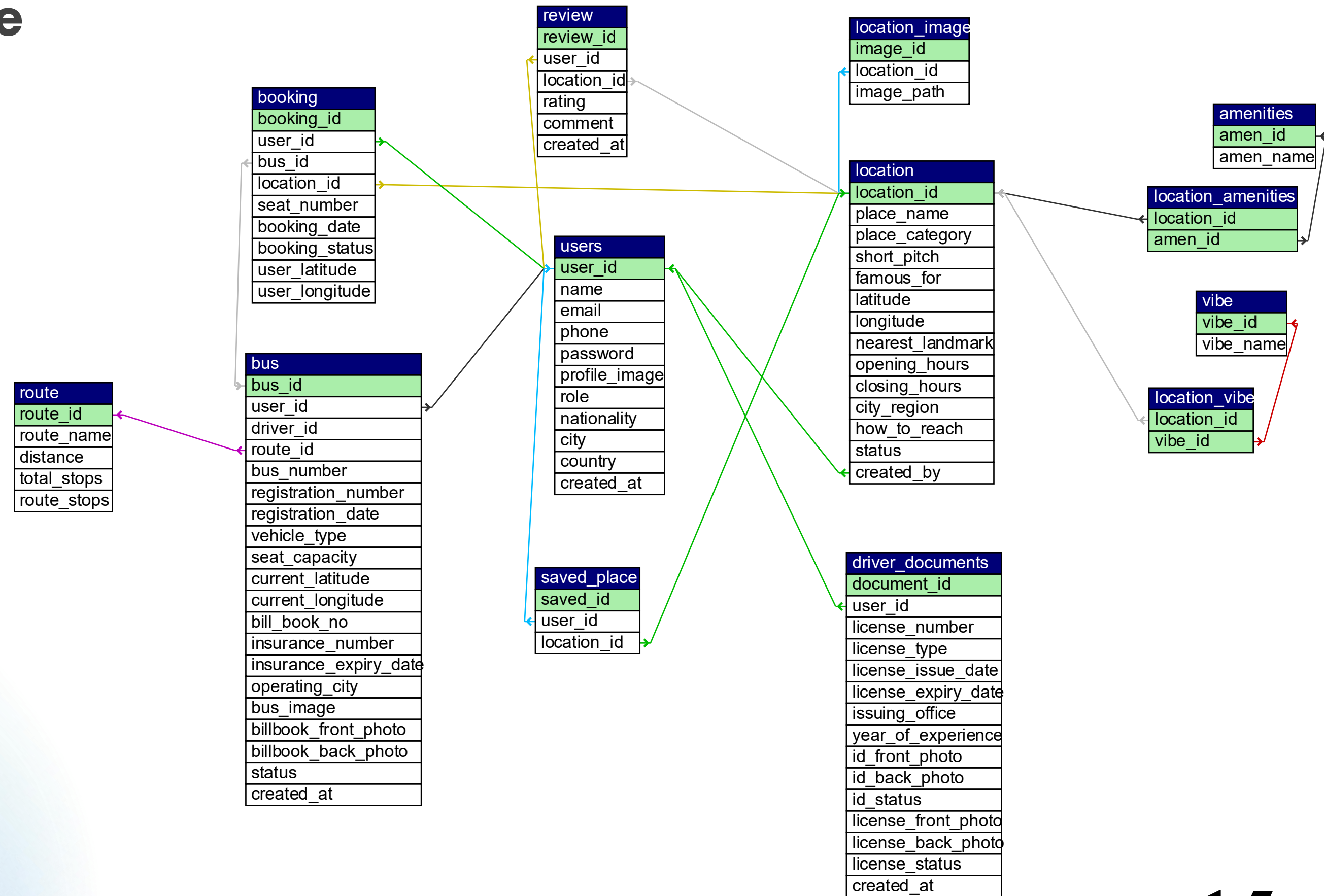




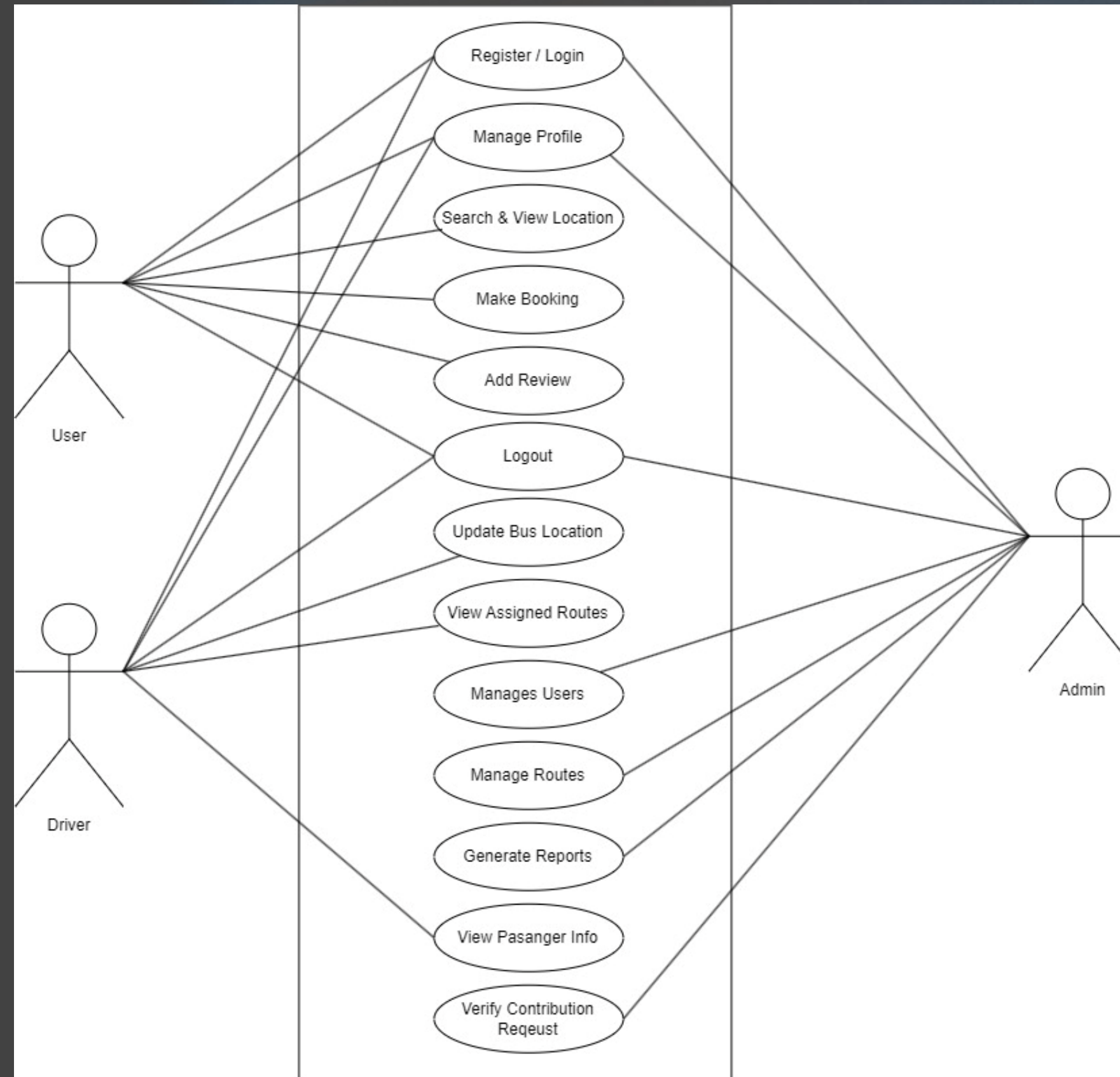
ER Diagram



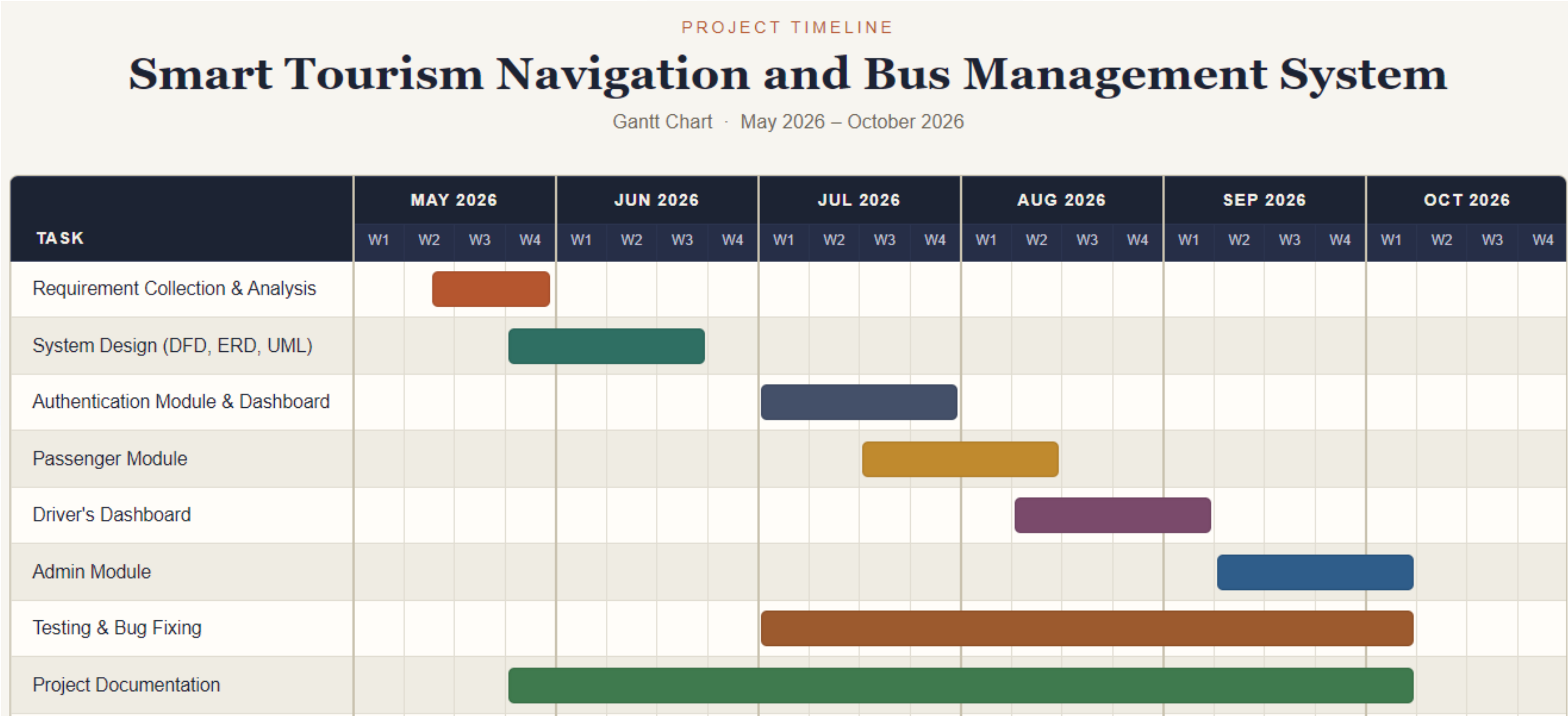
Schema Diagram



Use Case

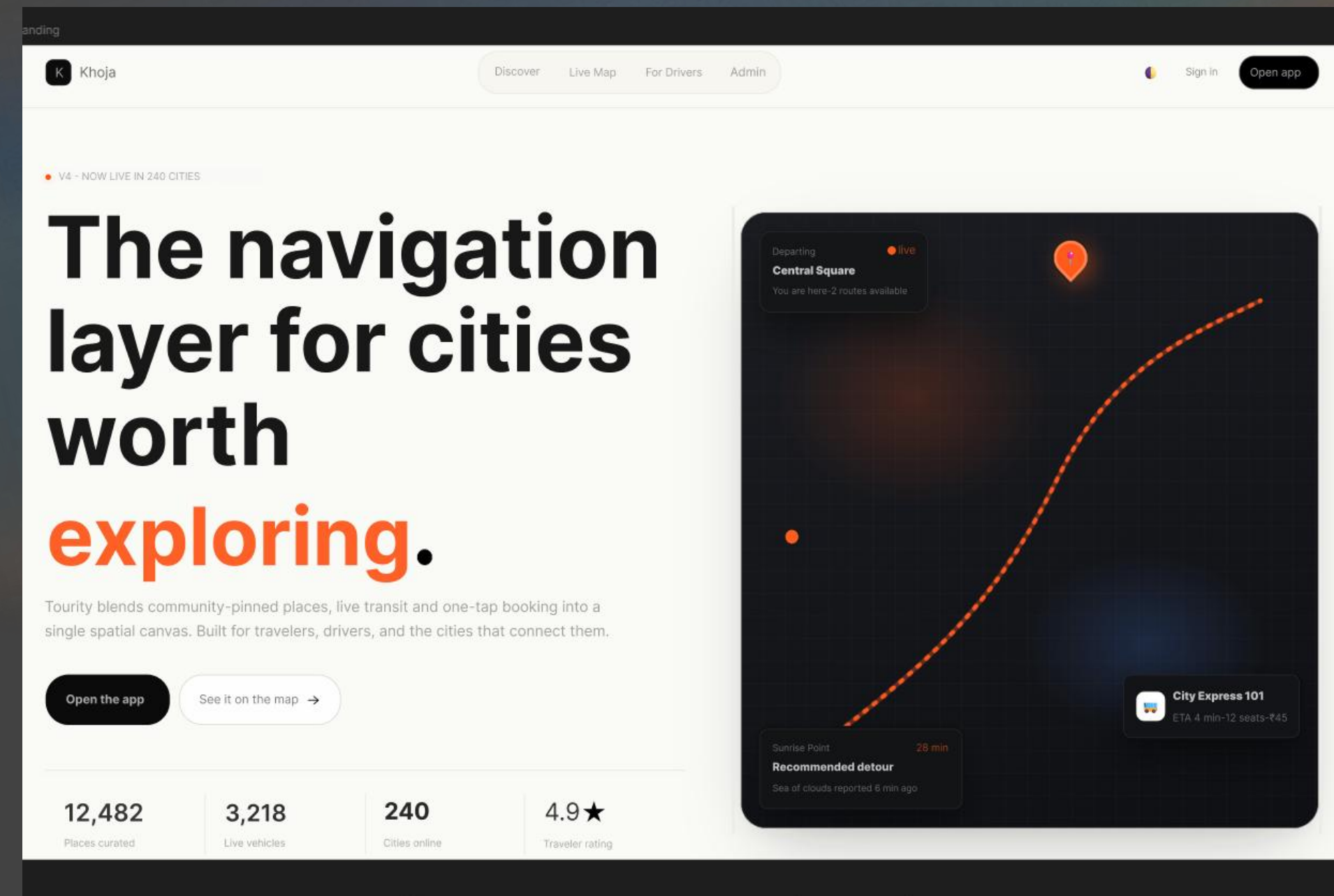


GANTT CHART

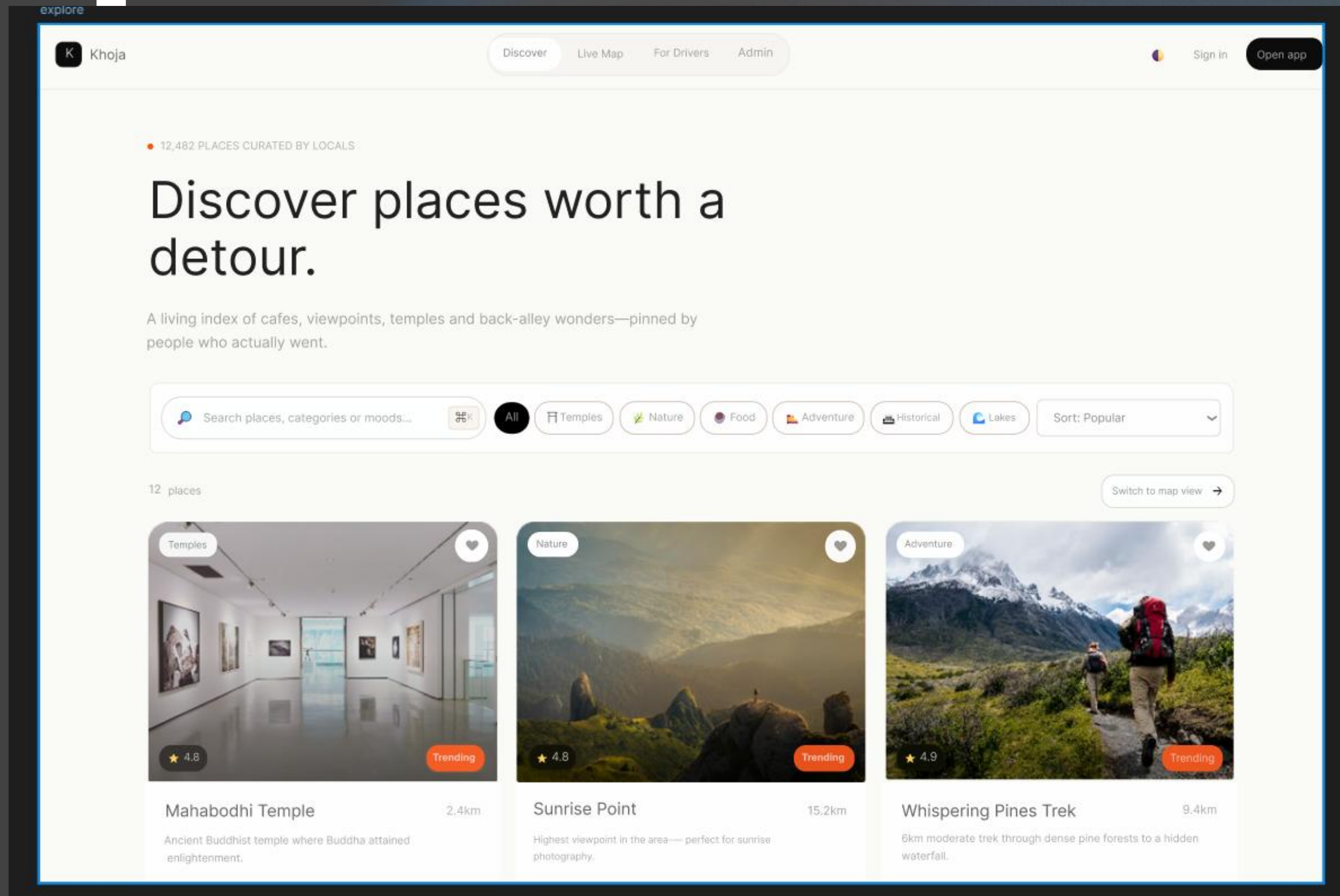


Expected Outcome

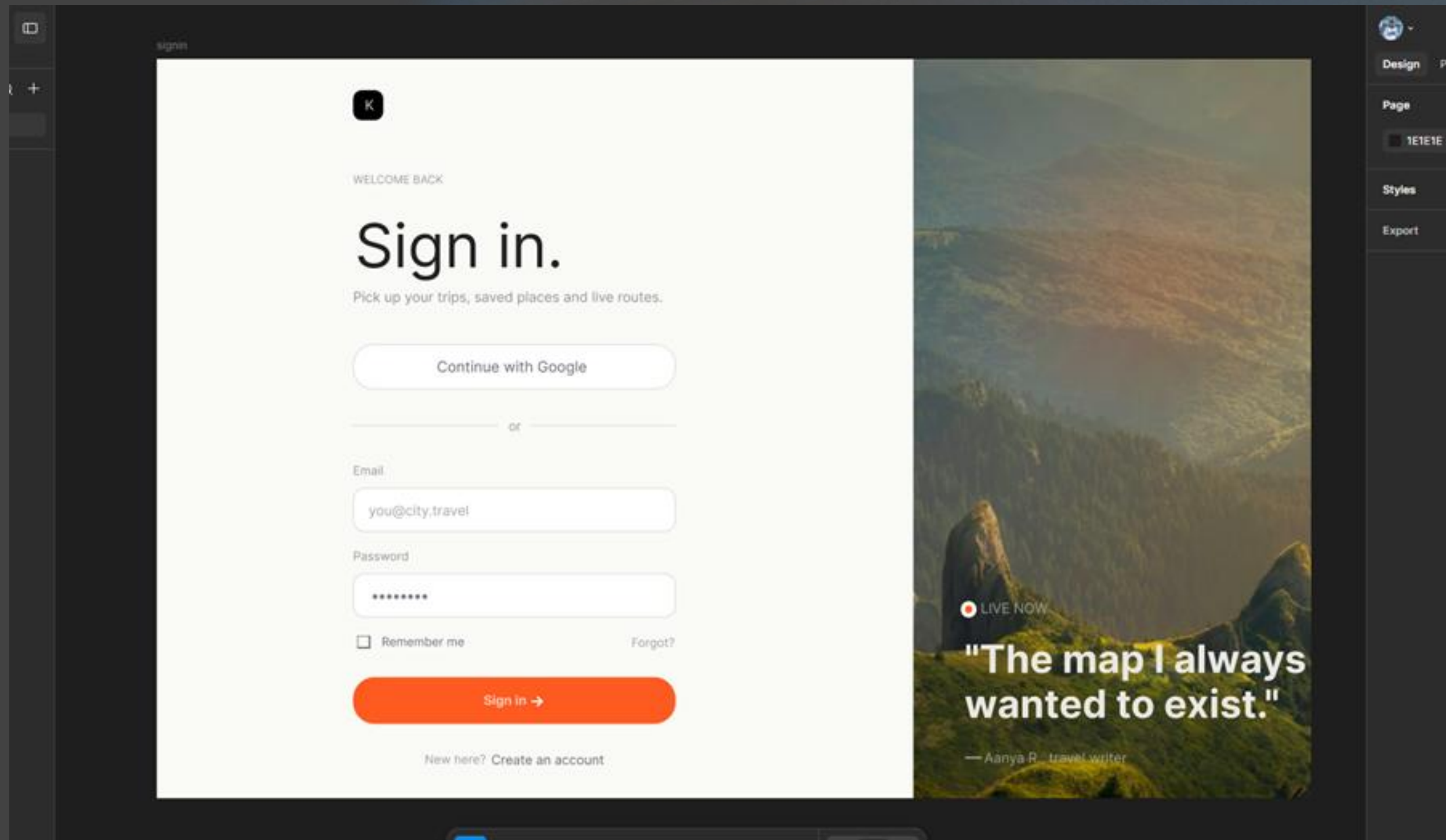
- A fully functional web-based platform featuring real-time bus tracking, GPS-based location discovery, and role-based dashboards.
- Increased public transport visibility and efficiency through a modern, responsive interface that supports community-driven mapping and visual route creation.



Expected Outcome



Expected Outcome



TESTING

Table 4.1: Test Case of unit testing for Login Module



SN	Test Case	Test Data	Expected Result	Status
1	User login with valid credentials	Valid email and password	User successfully logs in	Pass
2	Login with incorrect password	Valid email, wrong password	Error message is displayed	Pass
3	Login with empty fields	Email and password empty	Required-field validation is displayed	Pass
4	Driver login with valid credentials	Valid driver credentials	Driver dashboard is displayed	Pass
5	Admin login with valid credentials	Valid admin credentials	Admin dashboard is displayed	Pass

TESTING

Table 4.2: Test Case of unit testing for Map Module

SN	Test Case	Test Data	Expected Result	Result
1	Load interactive map	Valid map connection	OpenStreetMap map loads successfully	Pass
2	Display user's location	Valid GPS coordinates	User location is displayed	Pass
3	Zoom and pan map	User interaction	Map responds correctly	Pass
4	Load map without internet	No network connection	Map should provide an appropriate error/fallback	Pass

TESTING

Table 4.3: Test Case of unit testing for Location Module

SN	Test Case	Test Data	Expected Result	Result
1	Display tourist locations	Valid location records	Locations are displayed correctly	Pass
2	Display invalid location coordinates	Invalid latitude/longitude	System should reject invalid coordinates	Fail

TESTING

Table 4.4: Test Case of unit testing for Contribute Module

SN	Test Case	Test Data	Expected Result	Result
1	Submit new location	Valid location details	Contribution is submitted successfully	Pass
2	Submit location with image	Valid details and image	Location and image are submitted	Pass
3	Submit incomplete contribution	Missing required fields	Validation message is displayed	Pass
4	Submit unsupported image	Invalid file type	File should be rejected	Pass
5	Submit duplicate location	Existing location details	Duplicate contribution should be prevented	Fail

TESTING

Table 4.5: Test Case of unit testing for Admin Module

SN	Test Case	Test Data	Expected Result	Result
1	Manage contributions	Pending contribution	Admin can approve or reject contribution	Pass

Completed

Task

User
Authentication

Driver
Authentication

Admin
Authentication

Interactive Map

User Location

Tourist Location
Display

Location
Contribution

Admin
Contribution
Management

User Access
Control

Real Time Bus
Tracking

WebSocle
Communication

Remaining

Task

Bus Booking

Driver Document
Verification
Driver Booking
Management

Bus Route
Management

Multi Bus Route
Navigation

More than two bus
switching

Duplicate location
detection

User management

Profile
Management

Booking History

Searching

Thank You

for your time
and attention